

InsureStack / Pyq Mulligan

Deep Research Analysis

Products · Problem Areas · Underwriting AI vs. Workflow Automation · Pyq Mulligan Stack

Research date: May 2026

Company	Insure Stack (formerly Mulligan / Pyq AI)
Founded	Pyq AI founded ~2024; Mulligan launched Oct 2025; rebranded Apr 2026
Acquirer	Prestige Trucking Insurance (parent: CSMV Underwriters)
Backing	Y Combinator · WillowTree Ventures
Target market	Commercial insurance brokers (GL, commercial auto, workers' comp)
Core claim	Automates 90% of broker manual work; saves 8+ hrs/broker/week; 95% fewer policy review errors
Compliance	SOC 2 Type II
AMS integrations	Applied Epic, AMS360, EZLynx, Veruna

. The Origin Story: Pyq → Mulligan → InsureStack

The chain of events is very recent. **Pyq AI** — backed by Y Combinator and WillowTree Ventures and founded by Aman Raghuvanshi — launched a commercial insurance broker automation platform called **Mulligan** in October 2025. In April 2026, **Prestige Trucking Insurance** (a Florida-based retail brokerage and parent of CSMV Underwriters) acquired Mulligan and rebranded the whole thing as **Insure Stack**.

Prestige had been one of Mulligan's earliest customers, deploying the platform across its own retail brokerage operations before public launch. After seeing how Mulligan transformed its workflows, Prestige made the strategic decision to bring the technology fully in-house while continuing to distribute it to other brokers as well.

The new leadership team is Christina Torres (CEO), Daniel Hausman (COO), and David Zhang (CTO). Pyq founder Aman Raghuvanshi remains in an ongoing advisory role.

Key implication: InsureStack.ai is not a standalone AI startup — it is a brokerage-owned, YC-pedigreed automation platform that is simultaneously a product sold to competitors of its parent company. This creates a structural tension that runs through every strategic decision the company will make.

. The Product Suite

All InsureStack products sit at the **broker operations layer** — they automate the manual paperwork surrounding the insurance transaction, not the insurance decision itself. The known modules are:

Quote Stack

Pulls quotes directly from carrier portals without human hands. The AI logs into portals, fills applications, retrieves quotes, and aggregates them side-by-side for brokers. This is the flagship product.

Submission Assembly

Automates the assembly of structured submission packets sent to carriers. Takes raw client data and organises it into carrier-ready format, reducing manual entry and reformatting.

Policy Comparison

Side-by-side AI comparison of policy language, coverage terms, and exclusions across carriers. Designed to surface meaningful differences rather than simply lay documents adjacent to each other.

Proposal Generation

Produces branded, client-ready proposals from raw quote and policy data, eliminating the manual formatting and layout work traditionally done in Word or PowerPoint.

Commission Reconciliation

Reconciles commission statements against what is recorded in the agency management system (AMS). Syncs with Applied Epic, AMS360, EZLynx, and Veruna — one of the most underserved pain points in broker back-office operations.

Metric	Claimed Result
Hours saved per broker per week	8+ hours
Manual workflow automated	Up to 90%
Policy review error reduction	Up to 95%
Compliance certification	SOC 2 Type II

Source: InsureStack / Mulligan press releases and Y Combinator company profile

. Problem Areas in the Products

1. Carrier Portal Fragility — the Biggest Hidden Risk

Quote Stack works by automating interactions with carrier portals — effectively browser automation / UI scraping. This is technically brittle. Every time a carrier updates their portal UI, the automation breaks. At scale across dozens of carriers and potentially hundreds of brokerages, the maintenance burden is enormous and the failure modes can be invisible: a quote silently not retrieved, a field filled with stale data, or an application that gets stuck mid-submission. Real-time reliability is very hard to guarantee without continuous monitoring infrastructure and carrier-specific regression testing.

2. No Direct Carrier API Rails

The reason InsureStack uses portal scraping is that most carriers do not expose APIs to brokers. ACORD standards exist but adoption remains patchwork. This means Quote Stack's technical foundation is fragile by design — always one carrier portal redesign away from a silent failure. Until carriers open up API access or InsureStack negotiates direct integrations (expensive, time-consuming), this structural brittleness persists.

3. AMS Breadth vs. Depth

Supporting Applied Epic, AMS360, EZLynx, and Veruna covers a reasonable share of mid-market brokers — but there are 50+ AMS systems in active use. Smaller brokers frequently run NowCerts, HawkSoft, Genesis, QQ Catalyst, or others. The commission reconciliation module is only as useful as your AMS is supported, which narrows the addressable market considerably.

4. Policy Comparison Hallucination Risk

This is the highest-liability product in the suite. When AI compares coverage language and a broker relies on that comparison to advise a client, an AI error translates directly into Errors & Omissions (E&O) exposure. LLMs hallucinate insurance policy terms — they may misread endorsements, confuse exclusion language, or miss sub-limits. Even at 95% accuracy, 1-in-20 comparisons contains an error. In a high-volume brokerage that is a meaningful number of potentially wrong coverage recommendations. The platform needs robust citation back to exact policy text and explicit uncertainty flagging.

5. Competitive Data Trust Problem

Now that InsureStack is owned by Prestige Trucking Insurance — itself a competing retail brokerage — every other broker customer is routing their quoting data, client submissions, and carrier relationships through a competitor's technology. Whether or not Prestige maintains strict data governance, the perception of competitive exposure is a real sales obstacle. The roadmap will also naturally tend to favour Prestige's own lines (trucking, commercial auto) over what independent brokers need.

6. Scope Is Commercial Lines Only — Narrow Within That

The platform handles general liability, commercial auto, and workers' comp well. It has not expanded to E&S (excess and surplus), specialty lines, complex D&O/EPLI, or personal lines. The more complex the risk, the less useful process automation becomes — an underwriter still makes the judgment call on a \$50M property risk

regardless of how fast the submission was assembled.

7. Acquisition-Stage Product Uncertainty

The platform changed owners, leadership, and brand in April 2026. The engineering team is in transition. Product roadmap priorities will shift toward what serves Prestige's own brokerage operations first. Early Mulligan customers are now customers of a company with a materially different strategic agenda — a normal post-acquisition risk but a real one that slows enterprise sales cycles.

Underwriting AI vs. Workflow Automation: The Core Tech Divide

This is the most important strategic distinction when evaluating InsureStack and the broader insurtech space. The two categories are fundamentally different in technical difficulty, regulatory exposure, data requirements, and liability profile.

Dimension	Workflow Automation (What InsureStack does today)	Underwriting AI (Where InsureStack signals it is going)
Core task	Replace human process steps: form filling, document extraction, portal navigation, reconciliation	Predict risk quality, price coverage, select or decline submissions
Technical difficulty	Moderate — LLMs for document understanding + RPA for portal automation	Very high — actuarial modelling, proprietary loss data, adversarial selection dynamics
Data requirement	Public-facing carrier portals, policy PDFs, AMS exports	Proprietary historical loss data, claims records, exposure data — scarce and siloed
LLM hallucination risk	Moderate — errors are recoverable (wrong quote retrieved, wrong field filled)	High — errors create direct financial and legal liability (wrong coverage bound or declined)
Regulatory exposure	Low to moderate — SOC 2 compliance sufficient for current scope	High — NAIC AI model bulletin, state algorithmic bias rules, fair access to insurance requirements
ROI time to value	Fast — demonstrable in weeks (hours saved per broker)	Long — requires actuarial validation cycles, regulatory approval, back-testing
Who has done it	InsureStack, Relativity6, Transacto, and many point-solution tools	Coalition, Lemonade, Corvus — all spent 5–10 years and raised \$200M+ to get there

The LLM Accuracy Reality in Insurance

Academic research on LLMs applied to insurance tasks reveals two key data points relevant to InsureStack's current and planned products:

- LLMs hallucinate at rates of **11–15%** on finance-related tasks without specific mitigation. This is unacceptably high for policy comparison or underwriting decisioning.
- With adversarial self-critique architectures — where a critic agent systematically challenges the primary agent's conclusions — hallucination rates drop to **~3.8%** and decision accuracy rises from 92% to 96%. Still not sufficient for autonomous binding but viable for human-in-the-loop review workflows.

Implication for InsureStack: The policy comparison module sits at the highest-risk intersection of LLM limitations and liability exposure. Implementing adversarial self-critique and mandatory human sign-off before any coverage recommendation is acted upon is not optional — it is a product safety requirement.

The Strategic Wedge: Why Broker Workflow First

InsureStack's path from broker workflow automation toward underwriting intelligence is the logical long-term play — Prestige's MGA subsidiary (CSMV Underwriters) provides proprietary loss and exposure data that most pure-play automation tools lack. But the gap between 'we automate your quote retrieval' and 'we help carriers price risk better' is enormous in terms of data science infrastructure, actuarial validation, and regulatory compliance. The companies that have bridged it successfully spent 5–10 years and raised hundreds of millions of dollars to do so.

The Pyq Mulligan Workflow Automation Thesis — Assessed

What They Got Right

- **Correct problem identification:** The biggest friction in commercial insurance is not the coverage decision — it is the process around the coverage decision. A broker spending 8 hours a week manually pulling quotes and building proposals is a software-solvable productivity problem.
- **LLM moment was the unlock:** Document understanding good enough to parse carrier portals, PDF policies, and commission statements without custom OCR training for every document type became viable only with modern LLMs. Mulligan timed this correctly.
- **End-to-end chaining over point solutions:** The key product insight was to chain submission → quote → compare → propose → reconcile into a single workflow rather than selling five separate tools that brokers have to integrate themselves.
- **Commission reconciliation as a wedge:** Reconciling commission statements was genuinely underserved — most brokers still do this in spreadsheets — and it creates daily, recurring engagement with the platform that other workflow tools lack.

Residual Challenges the Thesis Has Not Solved

- **Portal scraping brittleness:** Relying on UI automation for quote retrieval is a maintenance-intensive architecture. It does not scale elegantly and creates reliability risk that directly undermines the platform's core value proposition.

- **Accuracy ceiling without carrier cooperation:** The platform cannot achieve the accuracy levels needed for truly autonomous operations without carriers exposing structured APIs. This requires industry-level cooperation that Mulligan alone cannot drive.
- **Post-acquisition trust deficit:** Selling to broker competitors of your parent company requires airtight data governance, transparent roadmap governance, and probably an independent advisory board. None of this has been publicly announced.
- **Narrow line coverage:** GL, commercial auto, and workers' comp are the commodity end of commercial insurance. Moving upmarket into E&S, specialty, or complex financial lines is a different product with different data, different carriers, and different broker workflows.

Bottom Line

InsureStack is a well-executed workflow automation layer for commercial insurance brokers, now carrying an interesting but unproven path toward underwriting intelligence via its MGA parent. The product quality and claimed metrics are real — the YC backing, early customer traction, and SOC 2 certification all indicate a serious product, not vaporware.

The structural risks — carrier portal fragility, competitive trust under brokerage ownership, LLM accuracy in high-stakes policy comparison, and the enormous gap to true underwriting AI — are genuine and underappreciated by most coverage of this company.

The underwriting AI ambition is the right long-term direction but represents a 3–5 year build and a fundamentally different company than what InsureStack is today. The near-term product is a broker productivity tool; the long-term aspiration is an AI-native MGA infrastructure platform. The credibility of the former does not automatically transfer to the latter.

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